

Decidability of head-normalization for RPSs

Pablo Barenbaum

2023-02-18

We suppose given a fixed signature Σ of function symbols with arities. First-order terms are formed inductively using variables and applied function symbols.

Definition 1 (Recursive program scheme). A *recursive program scheme* (RPS) is a finite set of first-order rewriting rules $\mathcal{R} = \{\ell_i \rightsquigarrow r_i \mid i \in I\}$ such that all the left-hand sides are of the form $f(\vec{x})$ where $\vec{x}$ is a sequence of pairwise distinct variables, and such that $\text{fv}(\ell_i) \supseteq \text{fv}(r_i)$ for all $i \in I$. In a term of the form $f(\vec{s})$ we say that f is the *head symbol*. A function symbol is $\mathcal{R}$ -*unknown* if it is the head symbol of the left-hand side of a rewriting rule, and it is $\mathcal{R}$ -*basic* otherwise. A function symbol f must not occur in the left-hand side of more than one rule for the set $\mathcal{R}$ to be an RPS.

Definition 2 (Head reduction, head normal form). If s, t are terms, we write $s \xrightarrow{h}_{\mathcal{R}} t$ to mean that s $\mathcal{R}$ -*head reduces* to t , according to the following axiom:

$$\frac{(f(\vec{x}) \rightsquigarrow r) \in \mathcal{R}}{f(\vec{s}) \xrightarrow{h}_{\mathcal{R}} r[\vec{x} := \vec{s}]}$$

where $s[x_1, \dots, x_n := t_1, \dots, t_n]$ denotes the simultaneous substitution of the occurrences of each x_i in s by t_i . Note that $\mathcal{R}$ -head reduction is not closed by compatibility under contexts. A term s is $\mathcal{R}$ -*head reducible* if there exists a term t such that $s \xrightarrow{h}_{\mathcal{R}} t$. A term s is *in $\mathcal{R}$ -hnf* if it is not $\mathcal{R}$ -head reducible. Equivalently, s is in $\mathcal{R}$ -hnf if it is either a variable or of the form $f(\vec{s})$ where f is a $\mathcal{R}$ -basic symbol. A term s *has a $\mathcal{R}$ -hnf* if $s \xrightarrow{h}_{\mathcal{R}}^* t$ for some t in $\mathcal{R}$ -hnf. A rewriting rule is $\mathcal{R}$ -*head reducible* if its right-hand side is a $\mathcal{R}$ -head reducible term, and $\mathcal{R}$ -*head irreducible* otherwise.

Lemma 3 (RPSs with no head irreducible rules diverge). If all rules in $\mathcal{R}$ are $\mathcal{R}$ -head reducible and s is $\mathcal{R}$ -head reducible, then s does not have a $\mathcal{R}$ -hnf.

Proof. Suppose that s has a $\mathcal{R}$ -hnf. Then $s \xrightarrow{h}_{\mathcal{R}}^* s'$ for some s' in $\mathcal{R}$ -hnf. The reduction sequence must contain at least one reduction step because s is not a $\mathcal{R}$ -hnf by hypothesis, so, distinguishing the last step, we have that $s \xrightarrow{h}_{\mathcal{R}}^* s'' \xrightarrow{h}_{\mathcal{R}} s'$ for some s'' . The last step $s'' \xrightarrow{h}_{\mathcal{R}} s'$ must be an instance of a rewriting rule $f(\vec{x}) \rightsquigarrow r$, so this step must be of the form $s'' = f(\vec{t}) \xrightarrow{h}_{\mathcal{R}} r[x := \vec{t}] = s'$. Moreover, by hypothesis, the rewriting rule $f(\vec{x}) \rightsquigarrow r$ is $\mathcal{R}$ -head reducible, which means that r must be of the form $r = g(\vec{r})$ with g an $\mathcal{R}$ -unknown symbol. Hence $s' = r[x := \vec{t}] = g(\vec{r}[x := \vec{t}])$ where g is $\mathcal{R}$ -unknown, which in turn means that s' is not a $\mathcal{R}$ -hnf, a contradiction. $\square$

Definition 4 (Collapse of an irreducible rule). Let $\rho = (f(\vec{x}) \rightsquigarrow r) \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. We define s^ρ as follows by induction on the structure of s :

$$\begin{aligned} x^\rho &\stackrel{\text{def}}{=} x \\ f(\vec{s})^\rho &\stackrel{\text{def}}{=} r^\rho[\vec{x} := \vec{s}^\rho] \\ g(\vec{s})^\rho &\stackrel{\text{def}}{=} g(\vec{s}^\rho) \quad \text{if } f \neq g \end{aligned}$$

where if $\vec{s} = (s_1, \dots, s_n)$ we write $\vec{s}^\rho \stackrel{\text{def}}{=} (s_1^\rho, \dots, s_n^\rho)$.

Furthermore, if $\mathcal{R} = \{\ell_i \rightsquigarrow r_i \mid i \in I\} \uplus \{\rho\}$, we define $\mathcal{R}^\rho \stackrel{\text{def}}{=} \{\ell_i \rightsquigarrow r_i^\rho \mid i \in I\}$.

Lemma 5 (Collapse commutes with substitution). Let $\rho \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. Then $s[\vec{y} := \vec{t}]^\rho = s^\rho[\vec{y} := \vec{t}^\rho]$.

Proof. The rule must be of the form $\rho = (f(\vec{x}) \rightsquigarrow r)$. Let $\vec{y} = (y_1, \dots, y_n)$ and $\vec{t} = (t_1, \dots, t_n)$. We proceed by induction on s :

1. Variable, $s = y_i$ for $1 \leq i \leq n$: Then $y_i[\vec{y} := \vec{t}]^\rho = t_i^\rho = y_i[\vec{y} := \vec{t}^\rho] = y_i^\rho[\vec{y} := \vec{t}^\rho]$.
2. Variable, $s = z \notin \{y_1, \dots, y_n\}$: Then $z[\vec{y} := \vec{t}]^\rho = z = z[\vec{y} := \vec{t}^\rho] = z^\rho[\vec{y} := \vec{t}^\rho]$.
3. Instance of the ρ rule, $s = f(\vec{s})$. Then $f(\vec{s})[\vec{y} := \vec{t}]^\rho = f(\vec{s}[\vec{y} := \vec{t}])^\rho = r^\rho[\vec{x} := \vec{s}[\vec{y} := \vec{t}]]^\rho \stackrel{(\text{IH})}{=} r^\rho[\vec{x} := \vec{s}^\rho[\vec{y} := \vec{t}^\rho]] = r^\rho[\vec{x} := \vec{s}^\rho][\vec{y} := \vec{t}^\rho] = f(\vec{s}^\rho)[\vec{y} := \vec{t}^\rho]$. To justify the fourth equation in the chain, observe that $\text{fv}(r) \subseteq \text{fv}(f(\vec{x}))$ because $\mathcal{R}$ is a RPS.
4. Otherwise, $s = g(\vec{s})$ with $g \neq f$. Then $g(\vec{s})[\vec{y} := \vec{t}]^\rho = g(\vec{s}[\vec{y} := \vec{t}])^\rho = g(\vec{s}[\vec{y} := \vec{t}])^\rho \stackrel{(\text{IH})}{=} g(\vec{s}^\rho[\vec{y} := \vec{t}^\rho]) = g(\vec{s}^\rho)[\vec{y} := \vec{t}^\rho]$.

□

Lemma 6 (Simulation). Let $\rho \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. If $s \xrightarrow{\mathcal{R}}^* t$ then $s^\rho \xrightarrow{\mathcal{R}^\rho}^* t^\rho$.

Proof. The rule must be of the form $\rho = (f(\vec{x}) \rightsquigarrow r)$. It suffices to show that if $s \xrightarrow{\mathcal{R}} t$ in one step, then $s^\rho \xrightarrow{\mathcal{R}^\rho} t^\rho$. We consider two cases, depending on whether the step $s \xrightarrow{\mathcal{R}} t$ is an instance of the ρ rule, or an instance of another rule.

1. Instance of ρ : Then the step is of the form $s = f(\vec{s}) \xrightarrow{\mathcal{R}} r[\vec{x} := \vec{s}] = t$ and $s^\rho = f(\vec{s})^\rho = r^\rho[\vec{x} := \vec{s}^\rho] \stackrel{(\text{Lem. 5})}{=} r^\rho[\vec{x} := \vec{s}^\rho]^\rho = t^\rho$.
2. Instance of a rule $\sigma \in \mathcal{R} \setminus \{\rho\}$: Then s is of the form $s = g(\vec{s})$ with $f \neq g$, there is a rule $\sigma = (g(\vec{y}) \rightsquigarrow u) \in \mathcal{R} \setminus \{\rho\}$, and the step is of the form $s = g(\vec{s}) \xrightarrow{\mathcal{R}} u[\vec{y} := \vec{s}] = t$. Moreover, there is a rule $(g(\vec{y}) \rightsquigarrow u^\rho) \in \mathcal{R}^\rho$, so we have that $s^\rho = g(\vec{s})^\rho = g(\vec{s}^\rho) \xrightarrow{\mathcal{R}^\rho} u^\rho[\vec{y} := \vec{s}^\rho] \stackrel{(\text{Lem. 5})}{=} u^\rho[\vec{y} := \vec{s}^\rho]^\rho = t^\rho$.

□

Lemma 7 (Reverse simulation). Let $\rho \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. If $s^\rho \xrightarrow{\mathcal{R}^\rho}^* t$ then there exists a term t_0 such that $s \xrightarrow{\mathcal{R}}^* t_0$ and $t = t_0^\rho$.

Proof. The rule must be of the form $\rho = (f(\vec{x}) \rightsquigarrow r)$. It suffices to show that if $s^\rho \xrightarrow{\mathcal{R}^\rho} t$ in one step, then there exists a term t_0 such that $s \xrightarrow{\mathcal{R}} t_0$ and $t = t_0^\rho$.

Note that the step $s^\rho \xrightarrow{h}_{\mathcal{R}^\rho} t$ must be derived by an instance of some rule $\sigma \in \mathcal{R}^\rho$. By definition of $\mathcal{R}^\rho$, this rule must be of the form $\sigma = (g(\vec{y}) \rightsquigarrow u^\rho) \in \mathcal{R}^\rho$ and it is obtained from a rule $\sigma' = (g(\vec{y}) \rightsquigarrow u) \in \mathcal{R} \setminus \{\rho\}$. Hence we have that s^ρ is of the form $s^\rho = g(\vec{p})$ and the step is of the form $s^\rho = g(\vec{p}) \xrightarrow{h}_{\mathcal{R}^\rho} u^\rho[\vec{y} := \vec{p}] = t$. By induction on s , we claim that there exists a term t_0 such that $s \xrightarrow{h}_{\mathcal{R}}^* t_0$ and $t = t_0^\rho$:

1. Variable, $s = z$: Impossible, as $s^\rho = z$ and there are no steps $s^\rho = z \xrightarrow{h}_{\mathcal{R}^\rho} t$.
2. Applied function symbol, with head f : Then s is of the form $s = f(\vec{s})$ and we know that $g(\vec{p}) = s^\rho = f(\vec{s})^\rho = r^\rho[\vec{x} := \vec{s}^\rho]$. Recall that r is the right-hand side of a $\mathcal{R}$ -head irreducible rule, so r cannot be headed by a $\mathcal{R}$ -unknown symbol. Equivalently, r can either be a variable or headed by a $\mathcal{R}$ -basic symbol. We consider these two as subcases:
 - 2.1 If r is a variable, say $r = z$, then $r^\rho = z$ and we know that $r^\rho[\vec{x} := \vec{s}^\rho] = g(\vec{p})$. This means that z must be one of the variables in the sequence $\vec{x}$, as otherwise we would have $r^\rho[\vec{x} := \vec{s}^\rho] = z \neq g(\vec{p})$. So assuming that $\vec{x} = (x_1, \dots, x_n)$ and $\vec{s} = (s_1, \dots, s_n)$, we must have that $z = x_i$ and that $g(\vec{p}) = r^\rho[\vec{x} := \vec{s}^\rho] = z[\vec{x} := \vec{s}^\rho] = x_i[\vec{x} := \vec{s}^\rho] = s_i^\rho$. Hence $s_i^\rho = g(\vec{p}) \xrightarrow{h}_{\mathcal{R}^\rho} t$. By IH, there exists a term t_0 such that $s_i \xrightarrow{h}_{\mathcal{R}}^* t_0$ and $t = t_0^\rho$. To conclude, observe that, using the rule ρ , we have $s = f(\vec{s}) \xrightarrow{h}_{\mathcal{R}} r[\vec{x} := \vec{s}] = z[\vec{x} := \vec{s}] = x_i[\vec{x} := \vec{s}] = s_i \xrightarrow{h}_{\mathcal{R}}^* t_0$.
 - 2.2 If r is headed by a $\mathcal{R}$ -basic symbol, we claim that this case is impossible. Indeed, if $r = h(\vec{r})$ with h a $\mathcal{R}$ -basic symbol, then we can obtain a contradiction, since $r^\rho[\vec{x} := \vec{s}^\rho] = h(\vec{r})[\vec{x} := \vec{s}^\rho] = h(\vec{r}[\vec{x} := \vec{s}^\rho]) \neq g(\vec{p})$. For the last inequality, observe that $h \neq g$ because h is a $\mathcal{R}$ -basic symbol whereas g is a $\mathcal{R}$ -unknown symbol.
3. Applied function symbol, with head g : Then s is of the form $s = g(\vec{s})$ and we know that $g(\vec{p}) = s^\rho = g(\vec{s})^\rho = g(\vec{s}^\rho)$. In particular, $\vec{p} = \vec{s}^\rho$. Then we can take $t_0 := u[y := \vec{s}]$. Note that indeed we have $s = g(\vec{s}) \xrightarrow{h}_{\mathcal{R}} u[y := \vec{s}] = t_0$, using the rule σ' , and that $t = u^\rho[\vec{y} := \vec{p}] = u^\rho[\vec{y} := \vec{s}^\rho] \stackrel{(\text{Lem. 5})}{=} u[\vec{y} := \vec{s}]^\rho = t_0^\rho$.
4. Applied function symbol, with head $h \notin \{f, g\}$: We claim that this case is impossible. Indeed, in this case s is of the form $s = h(\vec{s})$ and we know that $g(\vec{p}) = s^\rho = h(\vec{s})^\rho = h(\vec{s}^\rho)$, but this is impossible because $g \neq h$.

□

Lemma 8 (Collapse of head normal forms). Let $\rho \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. Then:

1. If s is in $\mathcal{R}$ -hnf, then s^ρ is in $\mathcal{R}^\rho$ -hnf.
2. If s^ρ is in $\mathcal{R}^\rho$ -hnf, there exists s' in $\mathcal{R}$ -hnf such that $s \xrightarrow{h}_{\mathcal{R}}^* s'$.

Proof. The rule must be of the form $\rho = (f(\vec{x}) \rightsquigarrow r)$. We prove each part separately:

1. By induction on s :
 - 1.1 Variable, $s = y$: Then $s^\rho = y^\rho = y$ is in $\mathcal{R}^\rho$ -hnf.
 - 1.2 Applied symbol, $s = g(\vec{s})$: Since s is in $\mathcal{R}$ -hnf, g must be a $\mathcal{R}$ -basic symbol. In particular, $g \neq f$. Hence $s^\rho = g(\vec{s})^\rho = g(\vec{s}^\rho)$ which is also headed by a $\mathcal{R}$ -basic symbol, and thus it is in $\mathcal{R}^\rho$ -hnf.

2. By induction on s :

- 2.1 Variable, $s = y$: Then taking $s' := y$ it is immediate to conclude that $s = y \xrightarrow{\mathcal{R}}^* y = s'$ (in zero steps), and that $s' = y$ is in $\mathcal{R}$ -hnf.
- 2.2 Applied function symbol, $s = g(\vec{s})$: We consider two subcases, depending on whether $g = f$ or not:
 - 2.2.1 If $g = f$, then we know that $s^\rho = f(\vec{s})^\rho = r^\rho[\vec{x} := \vec{s}^\rho]$ is in $\mathcal{R}^\rho$ -hnf. Recall that r is the right-hand side of a $\mathcal{R}$ -head irreducible rule, so r cannot be headed by a $\mathcal{R}$ -unknown symbol. Equivalently, r can either be a variable or headed by a $\mathcal{R}$ -basic symbol. We consider three further subcases, depending on whether r is one of the variables in $\vec{x}$, a variable not in $\vec{x}$, or headed by a $\mathcal{R}$ -basic symbol:
 - 2.2.1.1 If r is a variable in $\vec{x}$ then $\vec{x} = (x_1, \dots, x_n)$ and $r = x_i$ for some $i \in 1..n$. Moreover, $\vec{s} = (s_1, \dots, s_n)$. Then we have that $s^\rho = r^\rho[\vec{x} := \vec{s}^\rho] = x_i[\vec{x} := \vec{s}^\rho] = s_i^\rho$, which is in $\mathcal{R}^\rho$ -hnf by hypothesis. By IH, there exists s'_i in $\mathcal{R}$ -hnf such that $s_i \xrightarrow{\mathcal{R}}^* s'_i$. Taking $s' := s'_i$, it suffices to note that $s = f(\vec{s}) \xrightarrow{\mathcal{R}} r[\vec{x} := \vec{s}] = x_i[\vec{x} := \vec{s}] = s_i \xrightarrow{\mathcal{R}}^* s'_i = s'$.
 - 2.2.1.2 If r is a variable not in $\vec{x}$, then $r = z$. Taking $s' := z$, which is in $\mathcal{R}$ -hnf, it suffices to note that $s = f(\vec{s}) \xrightarrow{\mathcal{R}} r[\vec{x} := \vec{s}] = z = s'$.
 - 2.2.1.3 If r is headed by a $\mathcal{R}$ -basic symbol, then we have that $r = g(\vec{r})$, where g is a $\mathcal{R}$ -basic symbol. Taking $s' := g(\vec{r}[\vec{x} := \vec{s}])$, which is in $\mathcal{R}$ -hnf, it suffices to note that $s = f(\vec{s}) \xrightarrow{\mathcal{R}} r[\vec{x} := \vec{s}] = g(\vec{r})[\vec{x} := \vec{s}] = g(\vec{r}[\vec{x} := \vec{s}]) = s'$.
 - 2.2.2 If $g \neq f$, then $s = g(\vec{s})$ is already in $\mathcal{R}$ -hnf, so it suffices to take $s' := s$, noting that $s \xrightarrow{\mathcal{R}}^* s = s'$ (in zero steps).

□

Lemma 9 (Equivalence of head normalization after collapse). Let $\rho \in \mathcal{R}$ be $\mathcal{R}$ -head irreducible. Then s has a $\mathcal{R}$ -hnf if and only if s^ρ has a $\mathcal{R}^\rho$ -hnf.

Proof.

- ($\Rightarrow$) Suppose that s has a $\mathcal{R}$ -hnf. Then $s \xrightarrow{\mathcal{R}}^* t$, where t is in $\mathcal{R}$ -hnf. By Lem. 6 we have that $s^\rho \xrightarrow{\mathcal{R}^\rho}^* t^\rho$. Since t is in $\mathcal{R}$ -hnf, by Lem. 8 (1) we have that t^ρ is in $\mathcal{R}^\rho$ -hnf. Hence s^ρ has a $\mathcal{R}^\rho$ -hnf.
- ($\Leftarrow$) Suppose that s^ρ has a $\mathcal{R}^\rho$ -hnf. Then $s^\rho \xrightarrow{\mathcal{R}^\rho}^* t$, where t is in $\mathcal{R}^\rho$ -hnf. By Lem. 7 there exists a term t_0 such that $s \xrightarrow{\mathcal{R}}^* t_0$ and $t = t_0^\rho$. Since $t_0^\rho = t$ is in $\mathcal{R}^\rho$ -hnf, by Lem. 8 (2) there exists t'_0 in $\mathcal{R}$ -hnf such that $t_0 \xrightarrow{\mathcal{R}}^* t'_0$. Since $s \xrightarrow{\mathcal{R}}^* t_0 \xrightarrow{\mathcal{R}}^* t'_0$, we conclude that s has a $\mathcal{R}$ -hnf.

□

Theorem 10 (Head normalization is decidable for RPSs). The question of whether a term s has a $\mathcal{R}$ -hnf is decidable.

Proof. Let $\mathcal{R}$ be an RPS and s an arbitrary term. We proceed by induction on the number of rules of $\mathcal{R}$ and consider two cases, depending on whether s is a $\mathcal{R}$ -hnf or not, which is a decidable predicate:

1. If s is a $\mathcal{R}$ -hnf, it trivially *has* a $\mathcal{R}$ -hnf, so return ‘yes’.
2. If s is not a $\mathcal{R}$ -hnf, note that s is $\mathcal{R}$ -head reducible. We consider two subcases, depending on whether all the rules in $\mathcal{R}$ are $\mathcal{R}$ -head reducible or not:
 - 2.1 If all the rules in $\mathcal{R}$ are $\mathcal{R}$ -head reducible, then by Lem. 3 we know that s does not have a $\mathcal{R}$ -hnf, so return ‘no’.
 - 2.2 Otherwise, let $\rho \in \mathcal{R}$ be a $\mathcal{R}$ -head irreducible rule. Note by Lem. 9 that s has a $\mathcal{R}$ -hnf if and only if s^ρ has a $\mathcal{R}^\rho$ -hnf, so we may proceed recursively over $\mathcal{R}^\rho$ and s^ρ . $\square$